

Fuel cells trail turbines for this route

Four fuel cell chemistries compete for stationary power today — **PAFC, MCFC, SOFC and PEMFC** — led respectively by HyAxiom (formerly Doosan Fuel Cell), FuelCell Energy, Bloom Energy, and a fragmented PEMFC field (Plug Power, Ballard, Cummins/Accelera, Hyundai HTWO, Toshiba H2Rex). Only two commercial products anywhere are sold as **dedicated pure-hydrogen units** — HyAxiom's PureCell M400 Hydrogen (PAFC, 440 kW, ~50% electrical efficiency) and Bloom Energy's Hydrogen-Powered Energy Server (SOFC, 300 kW+, ~52% average/60% peak) — and neither has a multi-MW, long-duration, independently verified hydrogen-fuel operating record; most of the sector's deployed capacity, including South Korea's 800 MW–1 GW+ stationary fuel cell fleet (the world's largest), runs on natural gas reformed on-site, not delivered hydrogen ([FuelCell Energy platform page](#); [HyAxiom M400 Hydrogen datasheet](#); [Bloom Hydrogen Data Sheet](#)). The one confirmed multi-MW plant genuinely fed by hydrogen rather than reformed gas is Korea's 50 MW Daesan plant, and it runs on industrial by-product hydrogen, not imported or ammonia-cracked hydrogen ([PowerMag](#)). Singapore itself has already chosen technology for its two flagship hydrogen/ammonia power projects — the 600 MW Keppel Sakra Cogen plant and a 55–65 MW ammonia-fed pilot — and both are **gas turbines, not fuel cells** ([Keppel](#); [Ammonia Energy Association](#)). For the MYSGH2 concept — importing green ammonia to MMHE, cracking it to hydrogen, and piping that hydrogen to Singapore for power — this means fuel cells are technically plausible as an offtake-side generation technology (their electrical efficiencies of 41–65% LHV generally match or beat simple-cycle gas turbines) but sit behind gas turbines in commercial readiness for grid-scale, pure-hydrogen duty in this exact region, and no Malaysia or Singapore deployment of a hydrogen-fed fuel cell exists to de-risk the choice.

No fuel cell chemistry is purpose-built for imported, cracked hydrogen at MW scale

The four chemistries split cleanly by temperature, fuel tolerance and typical unit size. **PAFC** (150–220°C, platinum catalyst, high CO tolerance) is dominated by HyAxiom's PureCell M400 family — 440–460 kW building-block units, sold in NG, LPG/NG dual-fuel, and dedicated Hydrogen variants, with HyAxiom having shipped a cumulative ~330.9 MW across ~758 units historically ([BCC Research company profile](#)). Fuji Electric remains a much smaller, Japan-centric PAFC player at the 100 kW class (FP-100E/FP-100i), with no evidence of MW-class installations. **MCFC** (550–650°C, internal reforming, no precious metal) is essentially a one-company field: FuelCell Energy's SureSource platform, sold in modules from ~300 kW to 3.7 MW and arrayed into "parks" up to 58.8 MW (the Gyeonggi Green Energy plant in Hwaseong-si, South Korea, still under a FuelCell Energy long-term service agreement as of FY2025) ([FuelCell Energy FY2025 10-K](#)). Crucially, FuelCell Energy's currently-shipping carbonate line runs on natural gas, biogas, or NG/H₂ blends up to 40–50% H₂ — **not pure hydrogen**; a 100%-hydrogen product is described as "in development" via a separate solid oxide line, so the MCFC platform as sold today cannot directly consume MMHE's cracked hydrogen stream without blending it back down with natural gas ([FuelCell Energy blog](#)).

SOFC (600–1000°C) is led by Bloom Energy, the only vendor with a clearly commercialized, spec-sheet-level pure-hydrogen product: the Hydrogen-Powered Energy Server, rated 300 kW and above on ≥99.90%-purity hydrogen, at an average 52% (initial maximum 60%) electrical efficiency ([Bloom Hydrogen Data Sheet](#)).

Bloom's base commercial module is 325 kW, aggregated into MW-scale plants, with a fleet exceeding 400 MW cumulative in South Korea via SK ecoplant and additional deployments in Taiwan (10 MW, Unimicron) ([Bloom/SK ecoplant 500 MW agreement](#); [Bloom Taiwan installation](#)). Mitsubishi Power offers a fundamentally different product — a pressurized SOFC integrated with a micro gas turbine — demonstrated at 206 kW AC output and 50.2% measured thermal efficiency over a 4,000-hour continuous test, but documented only on hydrocarbon fuel, with a single Tokyo commercial installation (Marunouchi Building, 2018) ([MHI](#)). Ceres Power, Convion, SolydEra, Elcogen and Sunfire round out the SOFC field as licensors or sub-100 kW players; several (Convion C60, SolydEra) offer hydrogen as a configurable fuel option, but only at pilot/CHP scale.

PEMFC (60–80°C) is the most fragmented chemistry commercially, with no single dominant utility-scale OEM. Plug Power's GenSure HP (built from 125 kW ProGen engines, configured at 500 kW–1.5 MW) and Ballard's FCgen-XD/heavy-duty modules (70–405 kW net) target data-center and grid-support backup power but remain at pilot stage for MW-class deployment; Cummins/Accelera operates a 1 MW PEM plant on surplus hydrogen at a South Korean refinery; Hyundai's HTWO 100 kW generator is the most fully specified PEMFC unit found, running on ISO 14687-2/SAE J2719-spec hydrogen at 7–9 bar supply with ≥50% LHV efficiency ([HTWO Fuel Cell Solution PDF](#)). Panasonic's ENE-FARM and Toshiba's H2Rex (a genuinely pure-hydrogen PEFC, stacked in 100 kW modules up to a 500 kW array at Tanaka Kikinzoku's Shonan Plant) sit at sub-MW, distributed-generation scale. Because ISO 14687 caps ammonia at just 0.1 ppm for PEMFC-grade hydrogen, any residual NH₃ slip from imperfect cracking at MMHE is a first-order fuel-quality risk specifically for this chemistry, more so than for the higher-temperature PAFC/MCFC/SOFC options.

The OEM comparison matrix, with every figure's basis and gaps preserved

The table below consolidates the sourced specifications found across all four chemistries. Where the research notes could not confirm whether an efficiency figure is stated on an LHV or HHV basis, that is marked explicitly rather than assumed — this matters because the project's own reference data (CLAUDE.md §5) states hydrogen LHV ≈120 MJ/kg vs. HHV ≈142 MJ/kg, a ~18% swing that changes any efficiency number's apparent value depending on convention.

Chemistry	OEM / Product	Unit power range	Operating temp	Fuel flexibility (H ₂ purity)	Electrical efficiency	CAPEX \$/kW	Footprint	Maturity/status
PAFC	HyAxiom (Doosan) PureCell M400	440–460 kW/module	150–220°C	NG, LPG/NG, or dedicated Hydrogen model	NG: 43% (basis unstated); H ₂ : 50% (basis unstated) (datasheets)	Not found; DOE 2015/2020 planning targets only (\$3,000 → \$1,500/kW), not actual cost	Unit envelope 8.9×2.5×3.0 m (~22 m ² base); no installed-plant figure	Commercial, shipping; ~330.9 MW/758 units cumulative in Korea
PAFC	Fuji Electric FP-100E/FP-100i	100 kW class	150–220°C	NG (reformed); no hydrogen model found	40% explicitly LHV (OSTI)	Not found	3.8×2.2×2.5 m (~8.4 m ² base), 12 t weight	Commercial in Japan; no current (2026) confirmation found; no

Chemistry	OEM / Product	Unit power range	Operating temp	Fuel flexibility (H2 purity)	Electrical efficiency	CAPEX \$/kW	Footprint	Maturity/ status
								MW-class parks
MCFC	FuelCell Energy SureSource (1500/3000/4000)	1.25–3.7 MW/module, arrayed to 58.8 MW parks	550–650°C	NG, biogas, NG/H2 blend up to 40–50% H2 ; pure-H2 line still in development (separate SOFC product)	SureSource 4000: ~60% (basis unstated); once-through MCFC generically ~47% LHV, 70–75% with fuel recycling	Not found (2024–2026); historical DOE targets only	~10,000 ft ² / 3.7 MW (<0.25 acre, ~250 m ² / MW, derived not OEM-stated); 59 MW park ≈5.2 acres (≈0.088 acres/MW, vendor-linked, unverified)	Commercial, shipping, but financially distressed (2024 global restructuring, workforce cuts 13–17%, continuing net losses through FY2025/26 per SEC filings)
SOFC	Bloom Energy Energy Server (incl. Hydrogen model)	325 kW/ module (NG); 300 kW+ (H2, ≥99.90% purity)	600–1000°C (SOFC generic)	NG, biogas, blended H2, or pure H2	Lifetime avg 54% LHV (NG), up to 65% BOL; H2 model 52% avg/ 60% peak (Bloom Hydrogen Data Sheet)	Historical (2018) ~\$7,000–8,000/ kW; no current figure; \$0.099/ kWh op-cost cited, not CAPEX	~107 m ² / MW (third-party, unconfirmed against primary datasheet)	Commercial, >1.5 GW deployed globally; stack replacement disputed (vendor: >5 yr; short-seller claim: ~2 yr)
SOFC	Mitsubishi Power SOFC-MGT hybrid	200 kW-class demonstrated; concept scales to 250 kW+	Pressurized SOFC + micro gas turbine	Hydrocarbon fuel (hydrogen operation not documented)	50.2% measured (4,000-hr test); CHP to 65–73%; 55–70% projected for future GTCC integration (not yet demonstrated)	Not found	Not found	Single Tokyo installation (2018); largely R&D-adjacent vs. MHI's core turbine business
SOFC	Convion C50/ C60 (Elcogen stacks)	25–60 kW	~600–1000°C	NG, biogas, or hydrogen (configurable)	C50: 50–55%; C60: ~60%	Not found	Not found	Sub-100 kW CHP/pilot scale only
PEMFC	Plug Power GenSure HP	500 kW–1.5 MW (125 kW ProGen blocks)	60–80°C (generic)	Hydrogen (purity spec not published per-unit); generic PEMFC needs ISO 14687 ≥99.97%	Not found for this product	Not found	Not found	Pilot at one US data center; not yet scaled commercial
PEMFC				Hydrogen		Not found	Not found	

Chemistry	OEM / Product	Unit power range	Operating temp	Fuel flexibility (H2 purity)	Electrical efficiency	CAPEX \$/kW	Footprint	Maturity/ status
	Ballard FCgen-XD / FCgen-HPS	70–405 kW (stationary modules); FCgen-HPS stack 140 kW	60–80°C (generic)		FCgen-HPS: 52% LHV, BOL (stack-level)			Catalog product (2024 datasheets); no large reference plant found
PEMFC	Cummins/ Accelera (ex-Hydrogenics)	100 kW–1 MW	60–80°C (generic)	Hydrogen (incl. surplus/refinery H2)	Not found	Not found	Not found	1 MW unit operating in South Korea refinery; commercial
PEMFC	Hyundai HTWO generator	70 kW rated / 100 kW peak	60–80°C (generic)	ISO 14687-2/ SAE J2719, 7–9 bar	≥50% LHV	Not found	~19.5 m ² /MW for FC module alone (derived from dimensions; excludes BOP/PCS)	Commercially cataloged
PEMFC	Toshiba H2Rex	3.5–100 kW modules, arrayed to 500 kW	60–80°C (generic)	Pure hydrogen	Not found	Not found	Not found	Multiple named Japan installs (2020–2024); largest 500 kW
PEMFC	Panasonic ENE-FARM	200–700 W (residential)	60–80°C (generic)	NG/LPG (reformed on-site)	40–41% LHV electrical; 98–101% LHV total CHP	Not found	Not found	Mature, mass-market since 2000s

Two patterns cut across this table that matter for MYSGH2. First, **no OEM in any chemistry publishes a current (2024–2026), primary-source \$/kW installed cost** — every figure found is either a stale historical data point (Bloom's 2018 S-1 filing), an aspirational DOE/EU program target, or absent entirely; this is a genuine data gap, not an oversight, and any CAPEX assumption used in downstream LCOH modeling must be logged in memory.md as an assumption per CLAUDE.md §7. Second, **footprint data is almost as thin**: only FuelCell Energy (MCFC) and, weakly, Bloom Energy (SOFC) have any plant-level footprint figure, and even those are vendor-sourced or third-party-derived rather than independently verified by a body like NREL, DOE or IEA.

Efficiency and footprint against this project's gas turbine and gas engine baseline

Set against this project's existing gas turbine reference range (simple-cycle 35–46% LHV, CCGT ~55–63% LHV, footprint ~0.111–0.139 acres/MW or ~450–563 m²/MW) and reciprocating gas engines (~\$1,500/kW capital rule-of-thumb), fuel cells hold their own on paper but not decisively. Electrical efficiency across the sourced fuel cell figures spans roughly **41–65% LHV**, comfortably above simple-cycle turbines and

competitive with CCGT — Bloom's SOFC (54% lifetime average, up to 65% beginning-of-life) and FuelCell Energy's MCFC (~60%, basis unstated) both sit in or above the CCGT band, while PAFC (41–50%) and PEMFC (≥50%) trail somewhat but still beat simple-cycle turbines. The one directly comparable footprint figure — FuelCell Energy's 59 MW MCFC park at ~5.2 acres, or ≈0.088 acres/MW — is modestly more compact than this project's gas turbine CCGT range, consistent with the industry's "modular, no large rotating equipment" claim, but it is a single vendor-linked data point, not an independent siting study, and no comparable figure exists for PAFC, SOFC or PEMFC at multi-MW scale. On cost, fuel cells do **not** look advantaged: even the most favorable fuel cell figures (DOE's \$1,000/kW 2030 target, or an unverified \$800/kW industry rule of thumb) land in the same band as 2025 gas turbine capital costs (\$728–1,544/kW per Gas Turbine World) and reciprocating engines (~\$1,500/kW), while most sourced fuel cell CAPEX figures (€5,000–18,400/kW for various SOFC/PEMFC configurations) are considerably higher. Combined with the fact that no independent TRL ranking places hydrogen-fed fuel cells against ammonia/hydrogen turbines or engines on a single scale, the strongest available evidence for relative maturity is behavioral: **Singapore itself, when it had a real commercial decision to make, chose gas turbines for both its flagship hydrogen-ready plant (600 MW Keppel Sakra Cogen) and its ammonia-fed pilot (55–65 MW), not fuel cells** — a signal from the exact offtake market this project targets.

Singapore has fuel cell activity, but none of it burns imported hydrogen

Asia-Pacific is where the fuel cell industry's real commercial weight sits, and Singapore specifically has more fuel cell activity than the MYSGH2 concept might assume — just not the kind this project needs. **South Korea** hosts the world's largest stationary fuel cell fleet (reported at roughly 800 MW to over 1 GW as of 2025, figures not reconciled between sources), built almost entirely on natural-gas-fed PAFC/MCFC under a Renewable Portfolio Standard that gives fuel cells a favorable REC weighting — though that policy is itself now under revision, with a bill to scrap the RPS system advancing as of May 2026 ([pv magazine](#)). Within that fleet, only the **50 MW Daesan plant** (Hanwha/Doosan, PAFC, by-product hydrogen from a petrochemical complex) is confirmed as genuinely hydrogen-fed rather than NG-reformed; other large Korean "hydrogen fuel cell" plants (Shinincheon Bitdream at 78.96 MW, Bloom/SK ecoplant sites) carry hydrogen branding but their actual fuel source was not confirmed in the research. **Singapore** has one confirmed fuel cell deployment — SK ecoplant is bringing Bloom Energy SOFC units to a GDS data center, described as Bloom's first Singapore project — but it is natural-gas-fed, not hydrogen-fed, and its fuel-cell-specific MW rating was not disclosed ([Bloom Energy](#)). A second, more directly relevant data point is **DayOne's 20 MW Singapore data center**, which broke ground with an explicit 0.3 MW SOFC hydrogen proof-of-concept and a Sembcorp partnership tied to Singapore's Green Data Centre Roadmap ([Hydrogen Central](#)) — the closest thing found anywhere in the region to a hydrogen-fed SOFC pilot, but it is explicitly a proof-of-concept at sub-MW scale, not a commercial power plant. **Taiwan** has a confirmed but NG-fed Bloom Energy SOFC deployment (up to 10 MW at Unimicron). **Malaysia has no fuel cell power generation deployment of any kind, at any scale, on any fuel, in any of the research notes reviewed** — its hydrogen activity is confined to production/export ambitions (Sarawak's Kuching and Bintulu projects) and small mobility-refuelling pilots, not stationary power. This is worth stating plainly for this project's own siting: **no Malaysia deployment and no Singapore hydrogen-fed fuel cell deployment were found anywhere in this research** — the closest regional precedents (Daesan's by-product hydrogen, DayOne's 0.3 MW proof-of-concept) are both smaller in scale, different in feedstock origin (industrial by-product vs. ammonia-cracked/imported), or both, compared to what MYSGH2 would require.

Conclusion

The research converges on a clear, if unwelcome, finding for anyone hoping fuel cells offer a ready-made offtake technology for MMHE's cracked hydrogen: the commercial fuel cell industry has built its scale on natural gas, not hydrogen, and where it has touched real hydrogen it has been industrial by-product streams (Daesan) or small pilots (DayOne), never imported, ammonia-derived hydrogen delivered by pipeline at the multi-MW-plus scale this project envisions. That gap is not fatal — Bloom's SOFC and HyAxiom's PAFC both sell catalog pure-hydrogen products today, and fuel cell electrical efficiencies compare favorably with gas turbines on paper — but it means any MYSGH2 permutation that specifies fuel cells for Singapore-side power generation would be pioneering a fuel pathway, not adopting a proven one, in a region where the dominant market participant (Singapore's own EMA-backed projects) has already picked gas turbines for both its hydrogen-ready and ammonia-ready flagship plants. The most actionable next step is not more literature search but direct vendor engagement: CAPEX, footprint, and hydrogen-specific efficiency data are thin enough across every chemistry that a request-for-information from HyAxiom, FuelCell Energy, and Bloom Energy — the three vendors with actual pure- or near-pure-hydrogen products — would likely surface more decision-useful numbers than any further desk research, and any figures adopted in the interim should be logged in memory.md as explicit assumptions per this project's No-Fabrication Rule.